SPE-227338-MS

Mechanically Robust 3D-Printed Proxies for Permeability Measurement of Burrowed Strata

Salma Hamed and Hassan Eltom, Geosciences Department, College of Petroleum Engineering and Geosciences, King Fahd University of Petroleum 5 and Minerals, Dhahran, Saudi Arabia; Abduljamiu Amao, Center for Integrative Petroleum Research, College of Petroleum Engineering and Geosciences, King Fahd 7 University of Petroleum and Minerals, Dhahran, Saudi Arabia; Ammar ElHusseiny, Geosciences Department, College of Petroleum Engineering and Geosciences, King Fahd University of Petroleum 5 and Minerals, Dhahran, Saudi Arabia; Mohammed Zaki Al-Abdul-Jabbar, College of Petroleum Engineering & Geosciences Technology Center, Remote Sensing & Digital Services, DTV, King Fahd 7 University of Petroleum and Minerals, Dhahran, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Quantifying and accurately measuring bulk permeability in bioturbated strata remains a significant challenge, primarily due to the difficulty of obtaining representative samples that adequately capture connected networks controlling fluid flow. These representative volumes often exceed the laboratory-scale testing. Previously, permeability assessments for such large samples have relied on numerical simulations, where direct permeability measurements for these large burrowed strata remain challenging.

However, recent advancements in computed tomography (CT) imaging and 3D printing technologies have enabled high-resolution reconstruction of burrow networks, which can then be scaled down to laboratory-compatible sizes. Despite this progress, issues persist, particularly with 3D-printed replicas produced via fused deposition modeling (FDM). These models frequently fail under the confining pressures applied during permeability testing due to structural deficiencies and fluid leakage.

After the failure of ten FDM-printed replicas, this study identifies critical parameters for achieving mechanically robust replicas. Specifically, Polyethylene Terephthalate Glycol) PETG (filament demonstrated markedly superior load-bearing capacity and pressure tolerance compared to Polylactic Acid PLA, significantly minimizing sample failure. Furthermore, the use of 100% matrix infill was found to be crucial for maintaining structural integrity, despite the longer print times required. Finally, applying an epoxy surface coating effectively sealed inter-layer voids inherent to FDM fabrication, thereby preventing fluid leakage and enhancing mechanical integrity for the printed replica.

The successful printing of pressure-tolerant 3D replicas represents a major advancement in experimentally measuring the permeability of bioturbated strata. This breakthrough paves the way for a better understanding of bulk permeability within the burrowed strata.

Introduction

Bioturbation, the reworking of sediments by organisms, results in the creation of burrows—fossilized structures comprising tunnels, shafts, and chambers excavated by animals within unconsolidated substrate (Bromley, 1996; Pemberton et al., 1992; Meysman et al., Knaust 2018). These burrows significantly affect parameters that control fluid flow within sedimentary strata, such as bulk permeability (Baniak etal 2022), specially when these burrows are larger than 4 mm – this cateorgy of bioturbation called macrobioturbation acrroding to Knaust classification Knaust (2014), sometimes these macro bioturbation burrow size may reach upto 50 mm and 1 cm, Eltom 2020 found that these burrow to capture the connected network that controls the fluid flow need larger Representative Elementary Volume (REV), or sample size of 20 cm × 14.4 cm, such samples size are far beyond lab capabilities to measure the permeability. Previously, challenges associated with determining bulk permeability in burrowed strata have been addressed through numerical modeling and simulation studies, which utilize parameters derived from CT data and Multi-Point Statistics (Eltom, 2023). However, directly calculating permeability remains impossible.

In general, research on bulk permeability across varying scales—ranging from large representative elementary volumes (REVs) to smaller-scale measurements—remains scarce. While numerical modeling has been the primary approach for advancing research in this area, this study introduces a novel solution for direct measurement of bulk permeability using 3D printing technology. The approach aims to address three key objectives related to accurately assessing permeability in burrowed strata.

On the other hand, 3D printing has emerged as a powerful tool that enables controlled experimentation on rock proxies across various scales, from Nano-resolution lithography to centimeter-scale flow models. Studies focus on replicating key characteristics, such as porosity, permeability, fluid flow behavior, wettability, and microstructural heterogeneity. For example, (Ishutov et al., 2015) and(Li et al., 2021). (2021) used Polyjet and two-photon lithography, respectively, to create scaled Berea sandstone proxies, achieving close matches to natural porosity (17.85–20.3%) and permeability (~988–1286 mD). Fluid flow dynamics in carbonates (Head & Vanorio, 2016) and fracture-vug systems (Yang et al., 2020) were simulated using SLA-printed models, revealing the impacts of diagenesis (compaction and dissolution) and geometry on permeability.

Furthermore, 3D printing was used to understand macroscopic rock properties—bulk-scale behaviors like acoustic, fluid flow, and electrical responses. These studies bridge theoretical diagenetic models with physical experimentation, showing how 3D printing enables controlled investigation of bulk properties (e.g., permeability evolution, seismic responses) by replicating specific rock features or processes measurably, where the primary study objective is to accurately measure the permeability of borrowed strata through downsizing burrowed samples into smaller samples that are compatible with current lab capabilities and conduct direct permeability measurements via water core flooding.

Geological background

Samples for this study were collected from the exposed uppermost Hanifa Formation of the Upper Jurassic at Jabal Al-Abakkayn near Riyadh, Saudi Arabia. Hanifa formation overlies the Tuwaiq Mountain Formation and is overlain by the Jubaila Formation (Powers et al., 1966). These central Arabian carbonates were deposited on the southern Tethyan margin, approximately 10° south of the equator, during the Middle to Late Jurassic (Al-Husseini, 1997). Hanifa Formation is divided into two members: the Hawtah and Ulayyah members, both characterized by a recessive interval at the base and a resistive interval at the top. The upper part of the Hanifa Formation is characterized by the Glossifungites Ichnofacies, which is generally defined by sub-vertical burrows in firm grounds that exhibit two primary burrow styles: pipe frameworks and bio-texturing. This study focuses on these pipe framework intervals, as they represent potential permeability pathways critical for understanding the diagenetic and reservoir characteristics of the Hanifa Formation. A larger sample size was taken and then sub-sampled into 10 smaller samples of 4 inches in diameter and 8 inches in length (3).

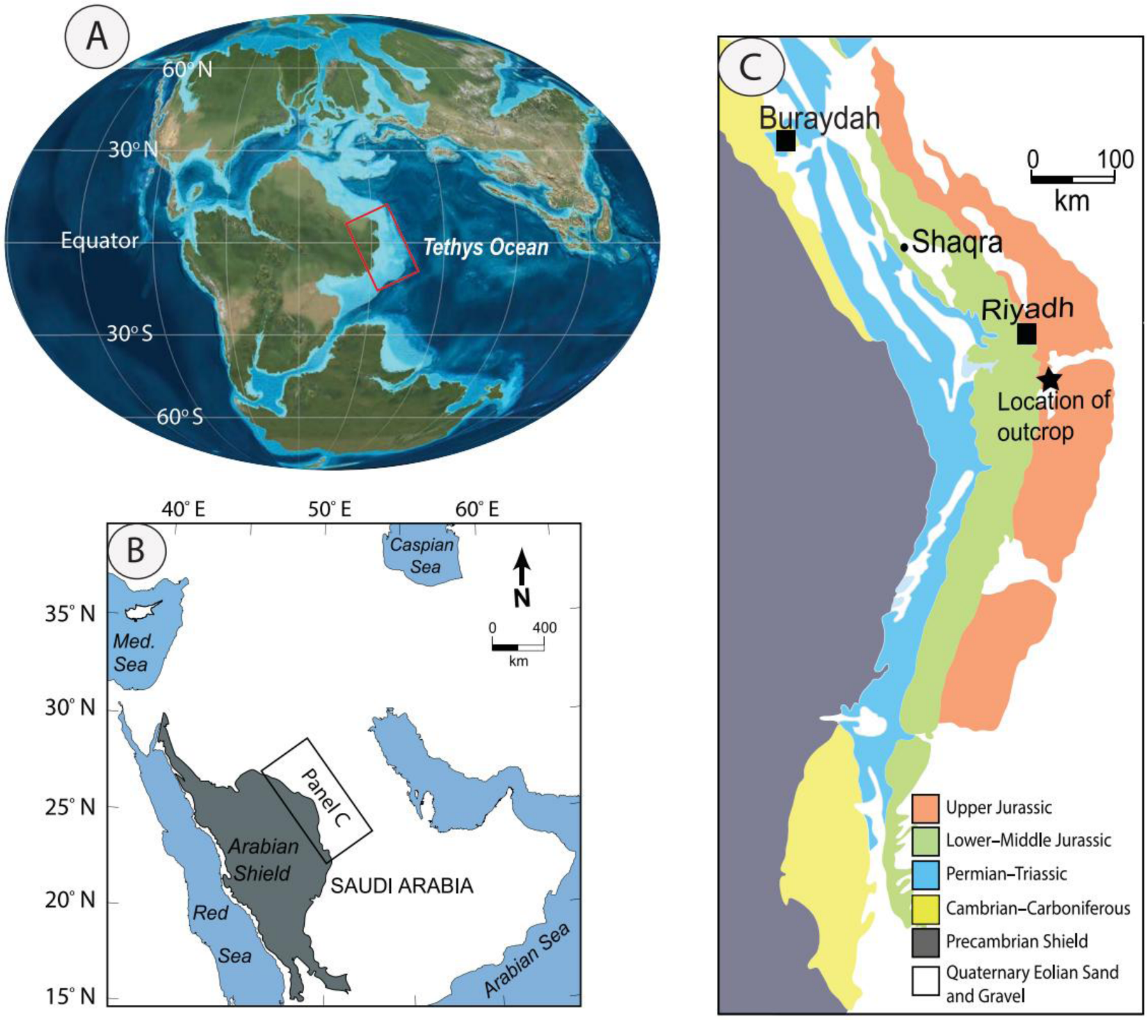

Figure 1—During the Middle to Late Jurassic, the Arabian Plate was positioned near the equator, approximately 10° south, along the southern margin of the Tethys Ocean, as shown in the paleogeographic map (courtesy of Ron Blakey). This positioning influenced the deposition of the Hanifa Formation.B) A regional map of the Arabian Peninsula highlights the central Arabian Shield and the adjacent Arabian Escarpment, providing context for the structural and sedimentary setting of the study area.C) A geological map (modified from Fischer et al., 2001) shows the distribution of the Arabian Shield and surrounding sedimentary rock units in central Saudi Arabia, highlighting the location of the Upper Hanifa Formation outcrops at Wadi Nisah, particularly the Alayah Member (Eltom, 2017).

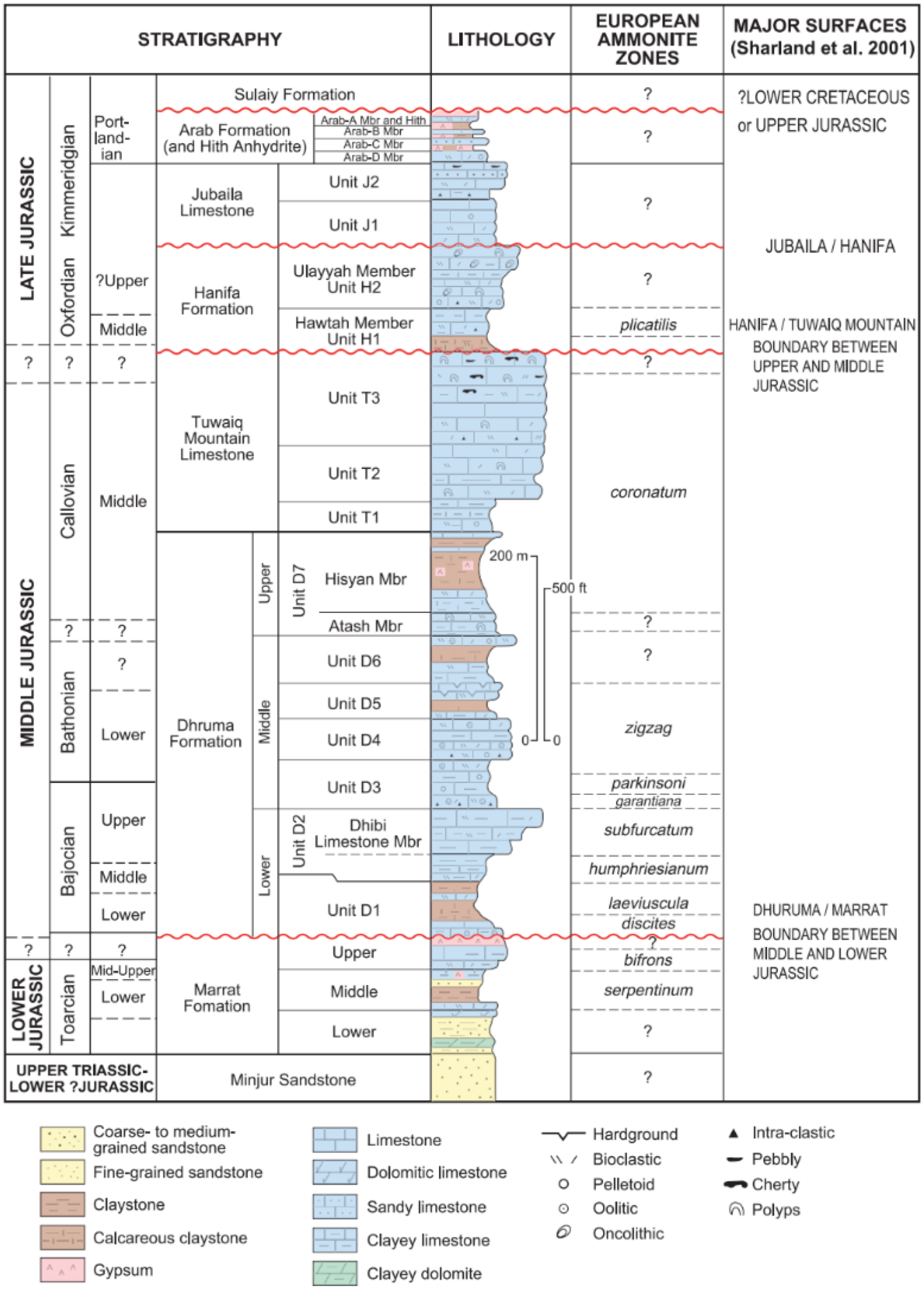

Figure 2—shows the stratigraphic position of the upper Jurassic of the Hanifia formation, in eastern Arabia Eltom, 2017

Methodology

Computed Tomography (CT) Imaging

Burrowed rock core samples (20 cm length, 5.1 cm radius; Figure 3) were scanned using a Toshiba Alexion scanner (TSX-032A) and VoxelCalc vX.0 software at the King Fahad University of Petroleum and Minerals CPG Center for Integrative Petroleum Research (CIPR) in 2022. Scans yielded images with a spatial resolution of 1 mm per voxel and a total tomographic volume of 14,787,500 voxels. From the ten scanned samples, sample (I) has the highest porosity fraction and the most interconnected burrow network, and was selected for subsequent analysis. This selection was performed because the primary purpose of this study is to investigate the permeability controls within a pre-existing connected pathway system.

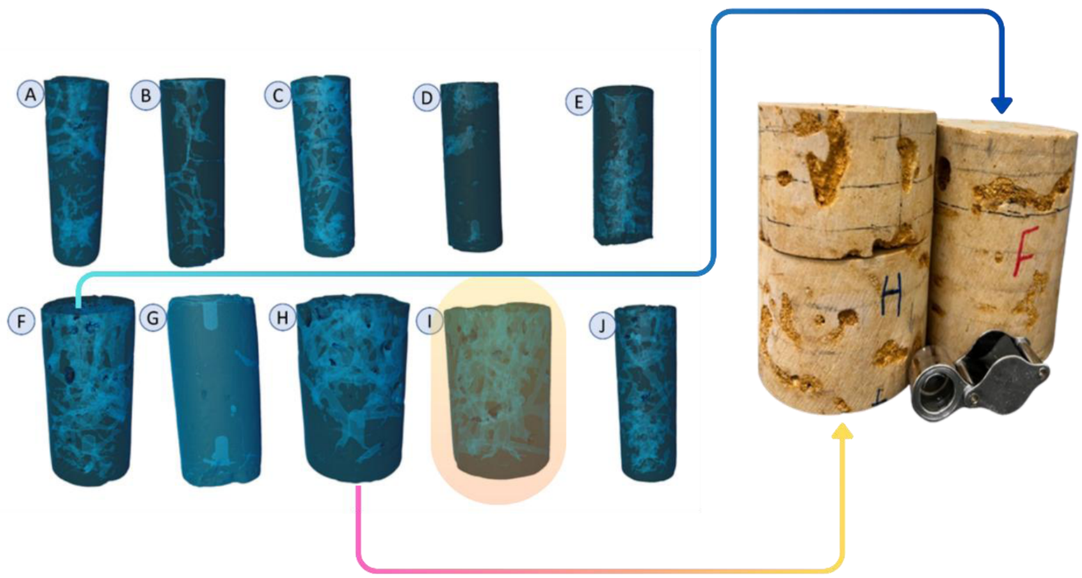

Figure 3—shows the CT images for the different collected samples, associated with the real rock sample for (H),(F) samples, while the (I) sample, which has the highest pore Volume, was chosen for this study.

Digital Rock Physics Modeling

Tomographic data were processed using Aviso 2022 software to generate a 3D-printable model, following a five-step workflow: 1) non-local mean filtering 2) Segmentation of burrow and matrix regions, 3) Pore network modeling and 4) porosity calculation 5) Iso-surfacing and triangular 3D meshing Initially, a non-local mean filter was applied to the tomographic data to smooth and de-noise the images before phase extraction. Although the data exhibited minimal acquisition artifacts, this filtering step was crucial for preserving edges and enhancing the definition of boundaries between the burrows and the surrounding matrix. In the next step, phase extraction was performed using the Edit New Label Field function within the Image Segmentation module to segment the tomographic volume into burrow and matrix phases. The matrix, which attenuates X-rays to a greater degree than air-filled burrows, appears as light gray or white in the tomographic data, while black or very dark gray regions represent burrows. This segmentation was achieved through intensity thresholding, where higher intensity values correspond to higher material density. The quantified pore volume obtained from this process provides a direct measure of the total porosity within each sample. Following segmentation. This was followed by the application of the Volume Fraction module to quantify the volumes of total pores, connected pores, and non-connected pores. Isolated pores were delineated by employing the Subtract Image module, which isolates disconnected pore volumes by subtracting connected pores from the total pore volume. For this study, the matrix was segmented, preserving the burrow network as negative space to facilitate its subsequent reproduction through 3D printing.

The matrix was converted into a mesh, a 3D surface representation composed of triangles that define the geometry of the matrix volume, with clearly defined interior and exterior regions (Ishutov et al., 2015). The resulting matrix mesh contained over 524,117 triangles, utilizing the fast and medium-quality meshing approach. Finally, the mesh was exported as an OBJ file of 128,445 KB for 3D printing.

3D Printing

Three-dimensional (3D) printing is a process of creating solid, three-dimensional objects from digital files. (Ishutov et al., 2015). It was used to produce physical replicas of the rock sample that had been burrowed. The digital object file format OBJ generated during the previous step was printed using a Raise3D E2 3D printer, utilizing Fused Deposition Modelling (FDM) technology. In FDM, a thermoplastic filament (such as PLA or PETG) is extruded through a heated nozzle, depositing molten material layer by layer onto a build platform. This technique enabled the precise construction of proxies that replicate the complex, burrowed geometry. To ensure these replicas possessed the mechanical robustness required for permeability experiments (core flooding), an iterative optimization of printing parameters and design was implemented, as follows.

Initial 3D Print (Original Scale). The model was printed at its original size (50-micron layer resolution) to accurately capture the burrow network geometry, with a resolution equivalent to that of a 1 mm CT voxel (Fig. 4).

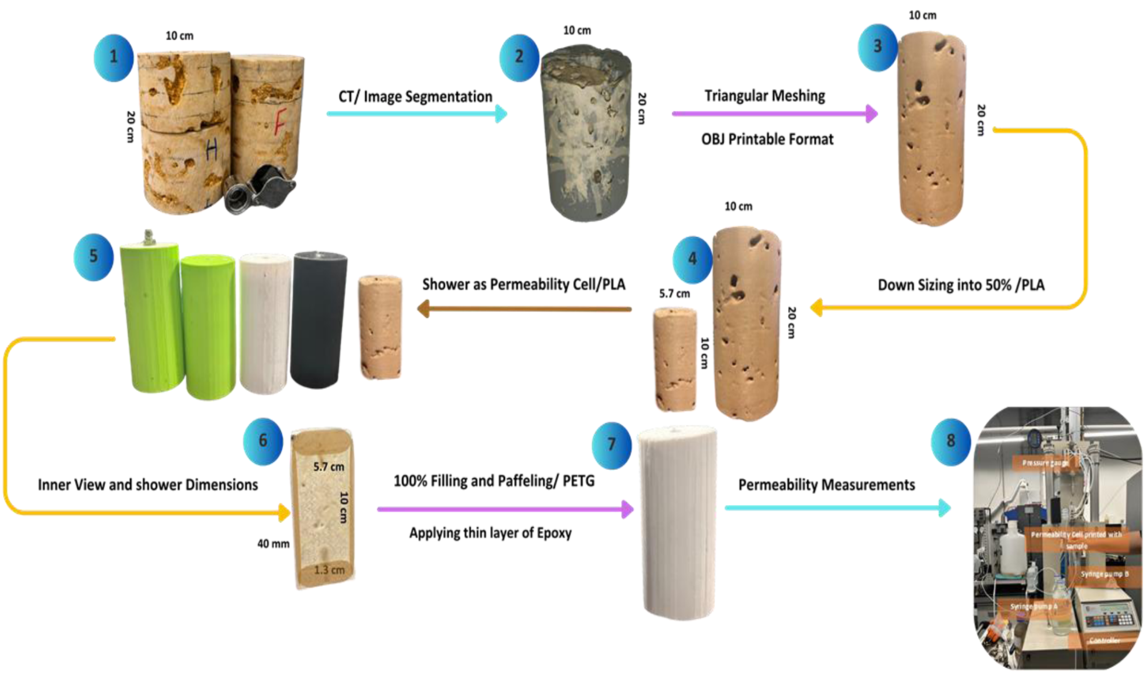

Figure 4—shows the steps and different stages from the real rock sample to printing replicas that can endure the pressure during permeability measurement.

Scaling Down (50% Size). In the second stage, the sample size was reduced to 50% of its original size, 50% of the original dimensions were printed with and without mesh infill using PLA material. Meshing reduced print time but critically compromised mechanical integrity, leading to sample failure (leakage) under permeability testing pressures, Fig. 4 (4).

Integrated Permeability Cell Design. in this step, an integrated flow cell (termed a "shower") was fabricated directly around a scaled-down sample replica using FDM and the same printing material (PLA/PETG), This monolithic design with a 40 mm thick lateral surround and 130 mm long extensions at the top and bottom Fig 4(6), This integrated design enables direct connection of syringe pumps and pressure gages to the cell's inlet/outlet extensions Fig 4(8), removing the need for complex setup. Consequently, permeability experiments could be conducted directly and efficiently using only a fluid pump and pressure monitoring system.

Solid Matrix and Material Optimization. In Stage matrix of (100% infill) was printed to enhance mechanical stability over a more extended period. Sample boundaries were rounded (adding baffles) to reduce stress. The printing material was changed from PLA to PETG due to its superior strength under core flooding pressures, as shown in Fig. 4 (7).

Micro-Porosity Sealing. The inter-layer gaps inherent to the FDM process create micro-porosity that acts as leakage pathways under experimental pressure. To eliminate this artifact, the PETG sample was coated with a thin layer of epoxy resin Fig. 4, 7). This treatment effectively sealed the micro-pores, concurrently enhancing the sample's mechanical strength and preventing fluid bypass during permeability measurements, ensuring flow was confined to the designed burrow network

Experimental permeability measurement

A single replica featuring an integrated permeability cell (5.7 cm diameter, 10 cm length) was successfully printed monolithically around the scaled sample. This assembly was connected directly to a syringe pump system (Fig. 5), enabling precise regulation of the fluid flow rate and continuous monitoring of pressure differentials. Permeability was determined by systematically varying the flow rate under steady-state conditions and recording the resultant pressure gradient. Absolute permeability was calculated by applying Darcy's law to the linear relationship between imposed flow rate and measured steady-state pressure differential.

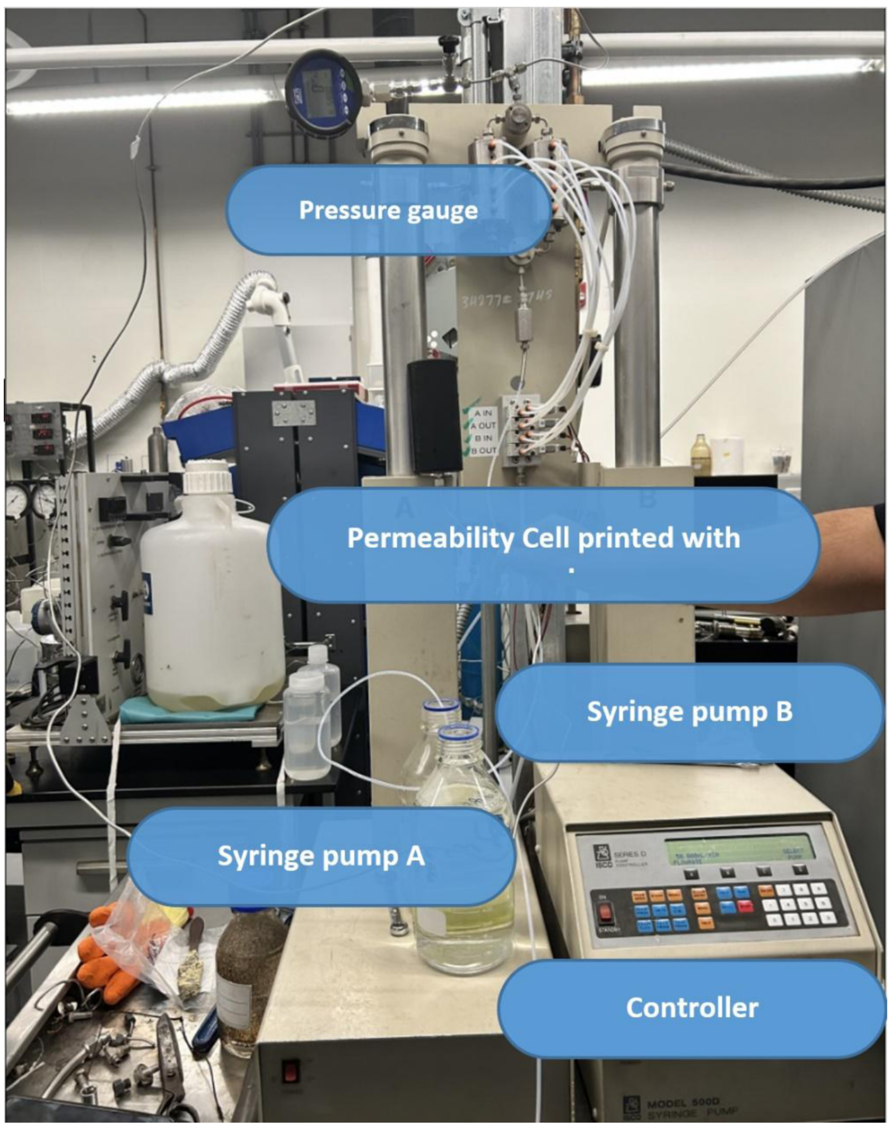

Figure 5—shows the experimental setup for measuring the permeability of the 3D printed replica.

3D Printing challenges for the burowed strata

Key challenges encountered during 3D printing proxy samples include:

Printing Duration

Printing solid matrix replicas is a time-intensive process. A scaled specimen (5.7 cm diameter × 10 cm length, 100% infill) required 72-80 hours. Printing at the original core dimensions would necessitate approximately 168 hours.

Mechanical Integrity

Replicas must withstand significant confining and pore pressures during core flooding experiments without deformation or failure.

Micro-Porosity

Inter-layer gaps inherent to FDM create micro-porosity, acting as unintended leakage pathways under pressure.

Permeability across the scale

Relating permeability measurements from scaled replicas back to the original rock sample dimensions.

Implications

This study introduces a novel experimental approach for replicating and investigating complex burrow networks, using mechanically robust 3D-printed proxies. The ability to physically reproduce burrow networks with high fidelity marks a significant advancement in experimental understanding of the bulk permeability of burrowed strata.

1. Novel Experimental Tool:
   The primary implication of this work is the demonstration of a cost-effective and scalable method for manufacturing and testing burrow-mimicking porous media. being able to produce these proxies will enable a controlled, repeatable approach for future experiments aimed at understanding how biogenic structures influence bulk permeability. This bridges the current gap between core-scale imaging and field-scale flow simulations.
2. Enabling Future Permeability Analysis:
   While direct permeability measurements were not presented in this study, the successful fabrication of mechanically strong replicas lays the foundation for future bulk permeability experiments. These will enable:
   - Systematic evaluation of how burrow parameters (e.g., orientation, connectivity, diameter, distribution) influence upscaled permeability.
   - Calibration and validation of digital rock physics and pore-network models using real-world-inspired, physically testable geometries.
3. This will also have Potential Applications in burrowed Reservoir Characterization or dual permeability reservoir in general:

Once permeability testing is carried out, this approach could support:

- Improved understanding of flow heterogeneity in dual permeability reservoir.
- Enhanced predictive capability for hydrocarbon recovery, groundwater flow, and subsurface energy storage.

- More accurate delineation of preferential flow pathways resulting from biogenic alteration.

In summary, while the current study does not yet quantify permeability, it establishes a validated fabrication framework that enables such investigations. This is a necessary and impactful step toward integrating burrow complexity into flow simulations and reservoir decision-making processes.

Conclusion

After fabricating 30 replicas, this study finds that the replicas withstand mechanical pressure during experimental permeability measurements. Firstly, PETG filament demonstrated significantly superior mechanical strength and pressure tolerance compared to PLA, substantially reducing sample failure during testing. Secondly, 100% matrix infill proved essential for maintaining structural integrity, despite requiring longer print durations. Finally, epoxy surface coating effectively eliminated leakage by sealing FDM-induced micro-porosity while concurrently enhancing mechanical robustness. Collectively, this integrated optimization, leveraging PETG material, solid infill, and epoxy sealing, enabled the successful measurement of permeability in 3D printed replicas under experimental pressures.

Acknowledgment

This research was conducted as part of my MSc thesis at King Fahd University of Petroleum and Minerals (KFUPM). I extend my sincere gratitude to the College of Petroleum Engineering and Geoscience (CPG) for providing cutting-edge facilities and resources essential to this work. We are thankful to Dr. Assad Barri for his expertise in guiding the permeability experiments. Special thanks to Hani Salman Al Mukainah for his invaluable contributions to CT image processing and 3D printing data preparation. My appreciation also extends to the entire Rock Physics Laboratory team at CPG for their technical support and collaborative spirit throughout this project.

Reference

Baniak, G. M., Gingras, M. K., & Pemberton, S. G. (2013). Reservoir characterization of burrow-associated dolomites in the Upper Devonian Wabamun Group, Pine Creek gas field, central Alberta, Canada. *Marine and Petroleum Geology*, **48**, 275–292. https://doi.org/10.1016/j.marpetgeo.2013.08.020

Baniak, G. M., La Croix, A. D., & Gingras, M. K. (2022). Recent advancements in characterizing permeability and porosity distributions in bioturbated flow media. *Earth-Science Reviews*, **232**, 104162. https://doi.org/10.1016/j.earscirev.2022.104162

Cunningham, K. J., Sukop, M. C., Huang, H., Alvarez, P. F., Curran, H. A., Renken, R. A., & Dixon, J. F. (2006). Prominence of ichnologically influenced macroporosity in the karst Biscayne aquifer: Stratiform "super-K" zones. *Geological Society of America Bulletin*, preprint(2008), 1. https://doi.org/10.1130/b26392.1

De Araújo, O. M. O., Aguilera, O., Coletti, G., Valencia, F. L., Buatois, L. A., & Lopes, R. (2021). X-ray micro-computed tomography of burrow-related porosity and permeability in shallow-marine equatorial carbonates: A case study from the Miocene Pirabas Formation, Brazil. *Marine and Petroleum Geology*, **127**, 104966. https://doi.org/10.1016/j.marpetgeo.2021.104966

Dey, J., & Sen, S. (2017). Impact of bioturbation on reservoir quality and production – A review. *Journal of the Geological Society of India*, **89**(4), 460–470. https://doi.org/10.1007/s12594-017-0629-4

Eltom, H. A., Alqubalee, A. M., Sultan, A. S., Barri, A. A., & Abdelbasit, K. (2021). Understanding the permeability of burrow-related gas reservoirs through integrated laboratory techniques. *Journal of Natural Gas Science and Engineering*, **90**, 103917. https://doi.org/10.1016/j.jngse.2021.103917

Eltom, H. A., & Goldstein, R. H. (2023). Digital rock modeling to quantify scale dependence of petrophysical measurements in burrowed reservoir rocks: An example using Thalassinoides. *Marine and Petroleum Geology*, **155**, 106412. https://doi.org/10.1016/j.marpetgeo.2023.106412

Eltom, H. A., González, L. A., Alqubalee, A., Amao, A. O., & Salih, M. (2020). Evidence for the development of a superpermeability flow zone by bioturbation in shallow marine strata, upper Jubaila Formation, central Saudi Arabia. *Marine and Petroleum Geology*, **120**, 104512. https://doi.org/10.1016/j.marpetgeo.2020.104512

La Croix, A. D., MacEachern, J. A., Ayranci, K., Hsieh, A., & Dashtgard, S. E. (2017). An ichnological-assemblage approach to reservoir heterogeneity assessment in bioturbated strata: Insights from the Lower Cretaceous Viking Formation, Alberta, Canada. *Marine and Petroleum Geology*, **86**, 636–654. https://doi.org/10.1016/j.marpetgeo.2017.06.024

Pemberton, S. G., & Gingras, M. K. (2005). Classification and characterization of biogenically enhanced permeability. *AAPGBulletin*, **89**(11), 1493–1517. https://doi.org/10.1306/07050504121

Uddin, Y. N., Clerke, E. A., Bammel, B., Koronfol, S., & Grader, A. (2017). Quantitative Study of Formation Vertical Permeability in Middle East Thalassinoides Burrowed Carbonate Mudstone. *One Perto*. https://doi.org/10.2118/186001-ms

Valencia, F. L., De Araújo, O. M. O., Buatois, L. A., Mángano, M. G., Valencia, G. L., Lopes, R., & Laya, J. C. (2023). Bioturbation changing porosity, permeability, and fracturability in chalk? Insights from an Upper Cretaceous chalk reservoir (Buda Formation, Texas, USA). *Ichnos/Ichnos: An International Journal for Plant and Animal Traces*, **31**(2–3), 176–198. https://doi.org/10.1080/10420940.2023.2212164

Valencia, F. L., De Araújo, O. M. O., Buatois, L. A., Mángano, M. G., Valencia, G. L., Lopes, R., & Laya, J. C. (2023). Bioturbation changing porosity, permeability, and fracturability in chalk? Insights from an Upper Cretaceous chalk reservoir (Buda Formation, Texas, USA). *Ichnos/Ichnos: An International Journal for Plant and Animal Traces*, **31**(2–3), 176–198. https://doi.org/10.1080/10420940.2023.2212164